

Anomalous Event Detection Based on YOLOv8 for Enhanced Security and Surveillance

Mohtasim Billah¹, Masud Raihan Leen², Md. Abu Ayaz Romim³, and Md. Saniat Rahman Zishan⁴

^{1,2,3,4}Department of Electrical & Electronic Engineering, Faculty of Engineering
American International University-Bangladesh(AIUB); 408/1, Kuratoli, Dhaka-1229, Bangladesh
shammo.4840@gmail.com, leenmasud2001@gmail.com, md.romim@gmail.com, saniat@aiub.edu

Abstract—This paper focuses on developing a multiple-camera, multiple-location monitoring enhanced security and surveillance system by integrating AI, particularly the machine learning algorithm YOLOv8 incorporated with the Deep Convolutional Neural Network (DCNN) model. The entire system consists of computer-integrated web cameras acting as CCTV cameras, IP cameras, and an autonomous rover for seamless surveillance. The whole system including the rover identifies objects like guns and knives, detects activities such as smoking, and responds effectively. With multiple cameras for real-time surveillance, the project achieves its goals, offering a unique fusion of machine learning and robotics for advanced security management. This research contributes to proactive surveillance systems, marking a notable advancement in security technology for safeguarding commercial infrastructures against diverse threats.

Keywords—YOLOv8, DCNN, Anomalous, CCTV Cameras, Autonomous Rover, LIDAR.

I. INTRODUCTION

In today's ever-changing world, safety and security have emerged as critical components of national development. People seek a sense of security and calm in their own country. But lack of security measures, thousands of human lives are being lost every day. In 2021, nearly 49,000 deaths in the United States were related to firearms [1] and at least 50 school shootings [2]. While our world has seen gains in security systems, there is still room for development, notably in activity recognition-related security and surveillance. A new solution has been created to address this issue: a cutting-edge, AI and Machine Learning-based activity recognition security system. This innovative security system intends to simplify and improve the monitoring process by seamlessly recording and automatically analyzing the real-time activity of persons in commercial areas with the help of the DCNN framework of the YOLOv8 algorithm. Its major goal is to improve safety and security by detecting objects and activities in multiple places in real time and discriminating between normal and possibly violent objects. The proposed autonomous rover can conduct real-time surveillance, continually patrolling specific regions and responding quickly to possible threats. This approach ensures an advanced and well-planned security procedure while maintaining the highest level of accuracy. Through autonomous navigation, the rover navigates the environment adeptly, surmounting obstacles, and optimizing its trajectory, thereby addressing crucial localization, mapping, and path planning challenges. The system efficiently processes vast volumes of CCTV/camera footage, employing advanced algorithms to extract pertinent information, track objects or individuals, and swiftly identify anomalies or potentially

hazardous activities. The project entails seamless integration of hardware components and software infrastructure while ensuring scalability, robustness, and ethical considerations suitable for commercial infrastructure applications.

II. LITERATURE REVIEW

A paper proposes a novel weapon detection (WD) model using the Pose Estimation and Long Short-Term Memory based Deep Convolutional Neural Network (PELSF-DCNN) framework to improve accuracy in surveillance systems. By combining You Only Look Once version 8 (YOLOv8) for object detection, the Diamond Search (DS) algorithm for motion estimation, and a sliding window process, the model effectively detects weapons from video frames. Key features are refined using the Cuckoo Search-Based Optimization (CSBO) algorithm and classified using PELSF-DCNN, achieving an accuracy of 97.5%, significantly outperforming existing models. The system addresses challenges like weapon occlusion, varied shapes, and false positives, though future work is needed to differentiate real and dummy weapons. The model demonstrates superior performance compared to traditional methods, making it a promising solution for weapon detection in security applications [3].

Research provides a unique approach for detecting unusual events in multi-camera surveillance videos. Unlike traditional methods, which treat cameras independently, this framework employs a coupled hidden Markov Model (CHMM) to combine information from various streams during inference. The approach involves a two-stage training process: first generating a set of usual events using clustering and user feedback, then training a precise CHMM model. Unusual events are identified by comparing the likelihood of test segments with the CHMM model [4].

A research focuses on anomalous human behavior identification, in video surveillance applications, and offers an update on earlier surveys. It lists the current approaches and draws attention to the most important problems facing the industry. Video data anomalous behavior has been detected using Bayesian modeling techniques such as Latent Dirichlet Allocation (LDA), Hierarchical Dirichlet Process (HDP), and probabilistic Latent Semantic Analysis (pLSA) [5].

A different study presents an unsupervised method for detecting unusual activity in videos using simple features. The approach involves segmenting the video and extracting features, then creating a prototype-segment co-occurrence matrix. The paper introduces an efficient algorithm for co-embedding and validates the approach through experiments on real-life videos. The proposed method contrasts with model-based approaches that require defining and modeling

"normal" activities, making it particularly useful for detecting unusual events in unconstrained environments [6].

A Research presented an intelligent and automatic system designed to efficiently monitor crowds and identify abnormal activities during Hajj and Umrah. A lightweight CNN model is trained on a custom dataset to identify pilgrims in surveillance footage. Another Slight weight CNN processes salient frames with pilgrims to extract spatial features. Utilizing a Long Short-Term Memory (LSTM) network, the system extracts temporal features from the processed frames. Results indicate promising accuracy rates of 81.05% and 98.00% on publicly available violent activity datasets, demonstrating the system's efficacy [7].

Another research offered a revolutionary neural network technique called "AnomalyNet" that incorporates these fundamental features into a cohesive framework for anomaly identification. This block is designed to keep the temporal and spatial links between motion and appearance signals intact. As a result, it successfully removes noisy backgrounds, collects motion-related information, and compensates for data inadequacies. It uses an adaptive iterative hard-thresholding method (adaptive ISTA), differing from typical sparse coding optimizers. This combination of 'L1-solver' and LSTM is a novel strategy that bridges the gap between these two approaches [8].

A deep neural network-based approach for identifying Chinese traffic signs has been suggested in a distinct study. It decreases computing complexity and achieves real-time detection. Experimental findings assessed on the extended Chinese traffic sign dataset and German Traffic Sign Detection Benchmark show that the suggested technique is more resilient and quicker than current methods, with a detection speed of 0.017 s per image. The study creates the CCTSDB dataset and suggests a real-time Chinese traffic sign identification system based on modified YOLOv2 [9].

III. METHODOLOGY & MODELLING

A. Block Diagram of the System

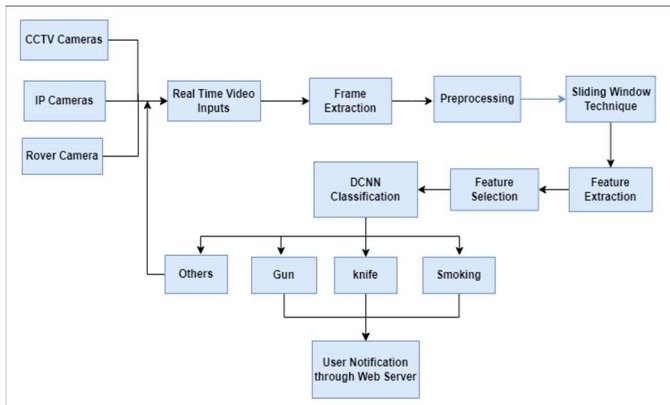


Fig.1. Block diagram of the system.

In Figure 1, The system begins by capturing real-time video inputs from various sources, including CCTV, IP, and rover cameras. These video streams are processed to extract individual frames, which serve as the basis for further analysis. Once extracted, the frames undergo preprocessing to enhance quality, making them suitable for detailed inspection. A sliding window technique is applied next, where a small window moves across the frame to focus on specific regions of interest. This is followed by feature extraction, where important visual elements, such as shapes

or textures, are identified. The system then performs feature selection, reducing the data by focusing only on the most relevant characteristics, streamlining the classification process.

The selected features are then passed into a Deep Convolutional Neural Network (DCNN). The DCNN is a powerful machine learning model designed to analyze visual data, especially in tasks like object recognition. It works by progressively learning patterns through multiple layers of convolution, pooling, and activation functions, which enable it to recognize complex objects such as guns, knives, or smoking activities. By processing the frame through these layers, the DCNN accurately classifies the content into specific categories, such as "Gun," "Knife," "Smoking," or "Others." Once classification is complete, the system sends real-time notifications to users via a web server if harmful objects or activities are detected, ensuring timely alerts and responses. Each stage of the system is interconnected, providing a smooth flow from video input to actionable detection and user alerts.

B. Flowchart of the Algorithm

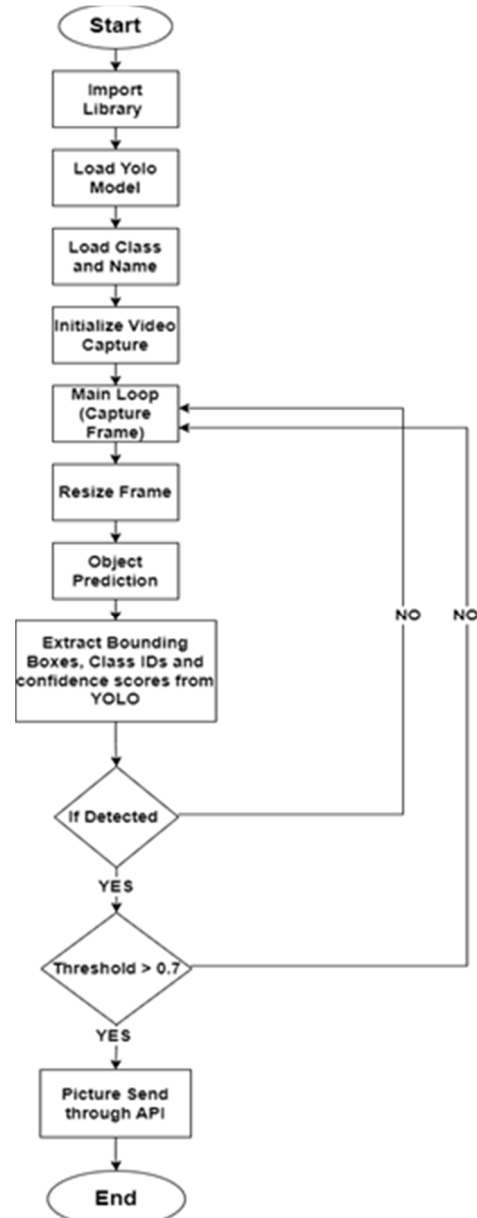


Fig.2. Flowchart of the algorithm.

This flowchart uses the YOLOv8 paradigm to show an object detection system. The pre-trained YOLO model and related class labels are loaded first, along with the essential libraries imported. After starting a video capture, every frame is resized to fit the input specifications of the algorithm. The object prediction model is applied to the resized frame, yielding bounding boxes, class IDs, and confidence ratings for items that are detected. The system determines whether the confidence score is higher than 0.7 when an object is detected. The identified image is delivered through an API for additional action if the threshold is satisfied. The procedure repeats to record and process the subsequent video frame if either no objects are found or the confidence level drops to less than 0.7. With real-time notifications based on high-confidence detections, the system offers an effective way for continuous item detection.

C. 3D Modelling of Autonomous Rover

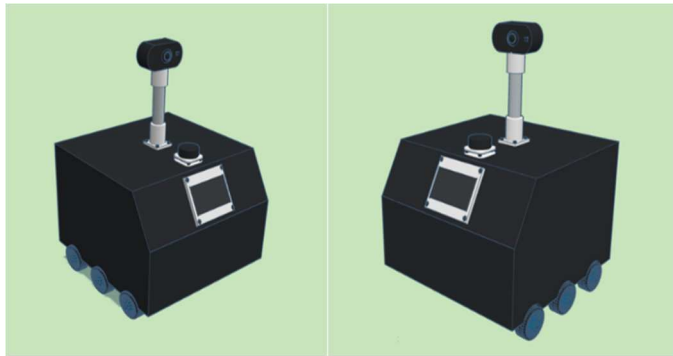


Fig.3. 3D Modeling of the rover.

The complete 3D model is precisely constructed, with precise measurements and realistic representations of each component. The Fusion360[10] software is used to build realistic 3D models of each component, allowing the entire rover system to be constructed. Overall, this 3D model shows the whole system with wheels, servo motors, a web camera with a stand, and a LIDAR that are all perfectly incorporated into an efficient and effective design.

IV. HARDWARE SETUP



Fig.4. Designed hardware model.



Fig.5. Raspberry Pi and 2D LiDAR setup.

In Figure 5, the Raspberry Pi facilitates autonomous rover navigation by interfacing with a 2D LiDAR sensor. Processing real-time data from the LiDAR, the Raspberry Pi enables the rover to make informed decisions, ensuring dynamic and obstacle-aware movement. This integration forms a robust system for precise and autonomous navigation in varied environments.

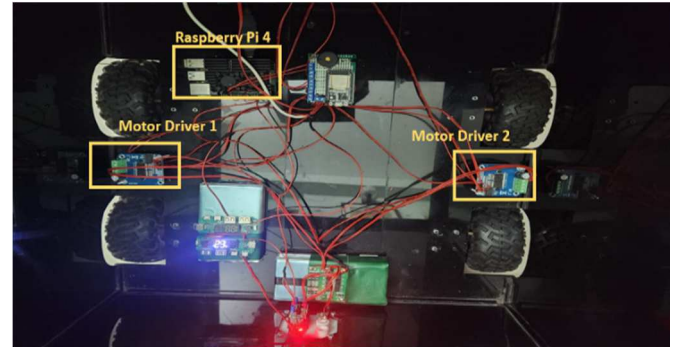


Fig.6. Motor Driver setup

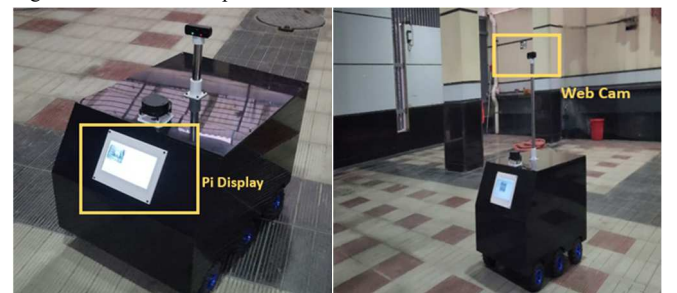


Fig.7. Pi Display and Web Cam setup.

In Figure 7, The AI program runs on the connected Pi display, utilizing the Raspberry Pi's processing power. The Raspberry Pi connects to an AI-equipped webcam for instant recognition of guns, knives, and smoking.

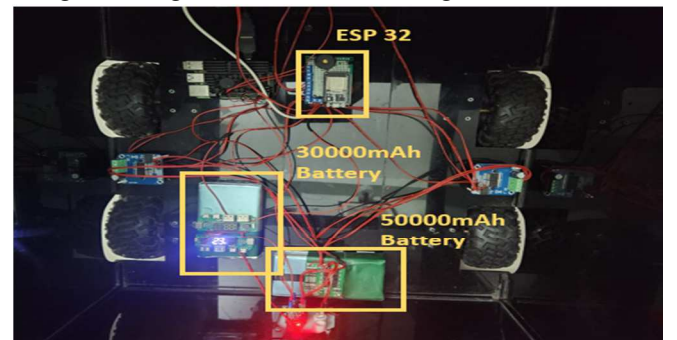


Fig.8. ESP32 and Battery setup.

In Figure 8, the rover operates on a dual-battery system, comprising a 30,000mAh and a 50,000mAh battery. This design ensures extended operational endurance, allowing the rover to perform autonomously for prolonged periods by efficiently managing power consumption with the combined capacity of both batteries.

V. SIMULATED RESULTS

The dynamic performance of real-time data captured by the camera holds the true simulation outcomes. It carefully sorts through this data, piecing together the anomalous fabric. The project underwent testing under various conditions, using different objects as models for the test cases.

A. Detection of Gun using a Single Camera



Fig.9. Detection of Gun using a Single Camera.

B. Detection of Knife using a Single Camera



Fig.10. Detection of Knife using a Single Camera

C. Detection of Smoking using a Single Camera



Fig.11. Detection of Smoking using a Single Camera.

D. Detection of Objects in Multiple Cameras

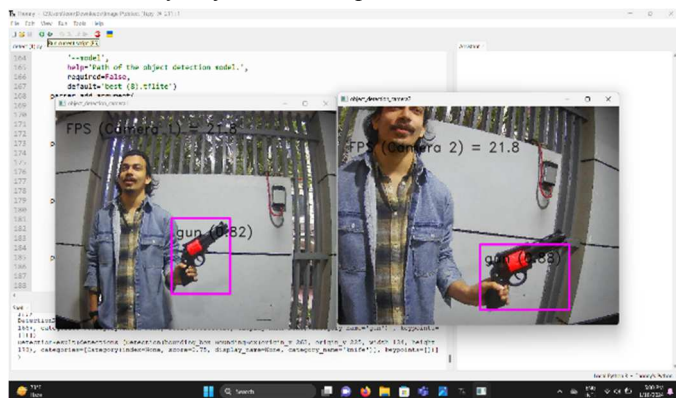


Fig.12. Detection of Gun using Multiple Cameras.

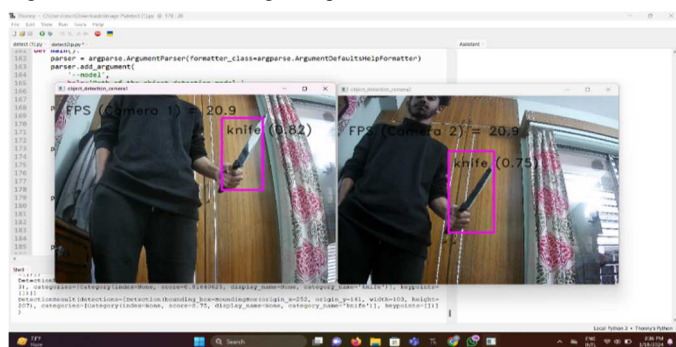


Fig.13. Detection of Knife using Multiple Cameras.

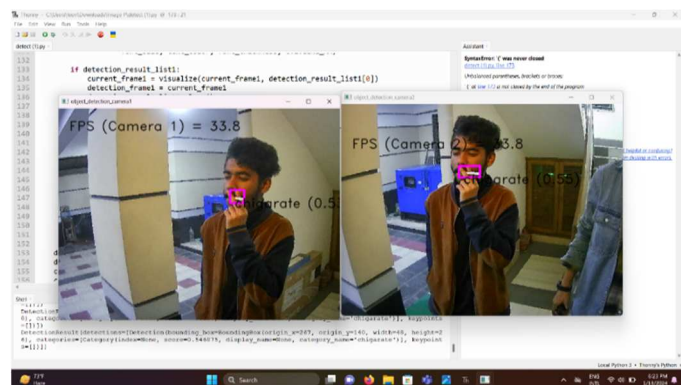


Fig.14. Detection of Smoking using Multiple Cameras.

In Figure 12,13,14, the capability of detecting anomalous objects in multiple cameras at a time has been shown. The results show accurate precision of detection.

E. Detection of Multiple Objects in a Single Camera

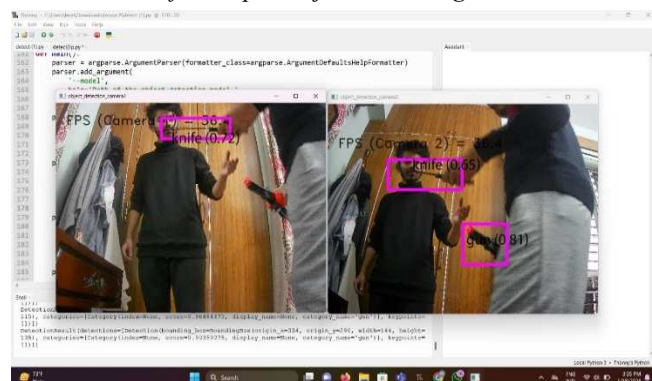


Fig.15. Detection of Multiple Objects in a Single Camera.

F. Detection of Multiple Objects in a Multiple Cameras

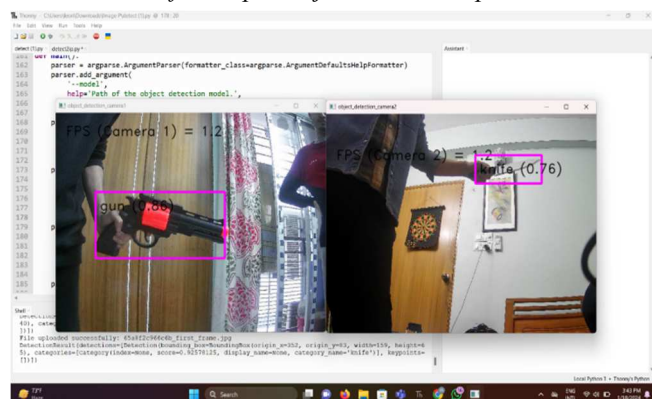


Fig.16. Detection of Multiple Objects in Multiple Cameras

G. Detection of Multiple Objects in Multiple Cameras in Multiple Locations at the same time

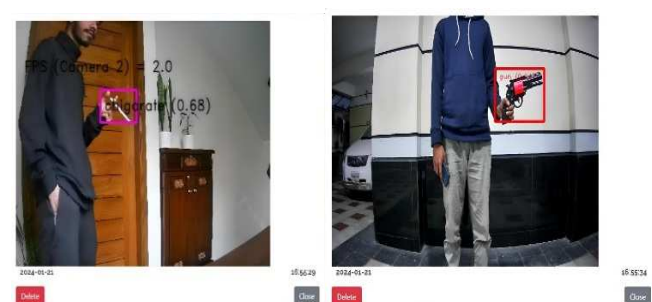


Fig.17. Detection of Multiple Objects in Multiple Cameras in Multiple Locations.

H. Detection of Objects in Multiple Type Cameras in Multiple Locations

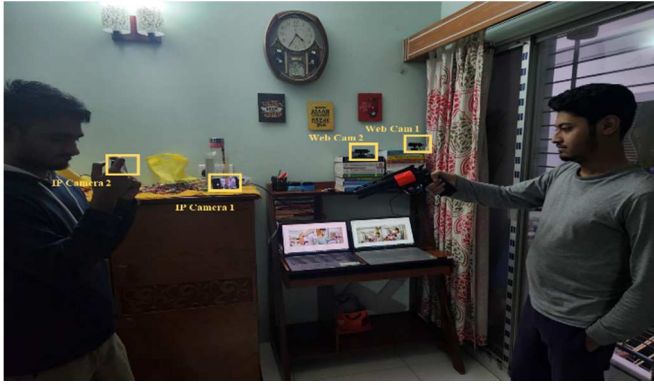


Fig.18. Detection of Objects in Multiple Type Cameras in Multiple Locations

VI. RESULT ANALYSIS

A. Distance Comparison with anomalous objects

TABLE I. OBJECT DETECTION DISTANCE COVERAGE BY CAMERAS

Distance Coverage (Meters)	Computer-integrated CCTV Cameras	Rover Camera	Mobile IP Cameras
0.1	Instant Detection	Instant Detection	Instant Detection
0.5	Instant Detection	Instant Detection	Instant Detection
1	Instant Detection	Instant Detection	Instant Detection
1.5	Instant Detection	Instant Detection	Instant Detection
2	Instant Detection	Instant Detection	Instant Detection
2.5	Instant Detection	Instant Detection	Instant Detection
3	Instant Detection	Instant Detection	Almost instant Detection
3.5	Instant Detection	Almost instant Detection	Almost instant Detection
4	Instant Detection	Almost instant Detection	Slight Delay Detection
4.5	Almost Delay Detection	Slight Delay Detection	Slight Delay Detection
5	Slight Delay Detection	Slight Delay Detection	Delay Detection

B. Multiple anomalous object detection comparison:

TABLE II. SUCCESS RATE OF DIFFERENT SYSTEMS DETECTING MULTIPLE ANOMALOUS OBJECTS

Numbers Of Anomalous Objects	Computer-integrated CCTV Camera	Rover Camera	Mobile IP Camera
One	100%	100%	100%
Two	100%	100%	66.67%
Three	100%	66.67%	33.33%

C. Accuracy Comparison between Cameras

TABLE III. COMPARISON OF OBJECT DETECTION ACCURACY

System	No of Test	Detected	Failed	Accuracy (%)
Computer-integrated CCTV Camera	50	48	2	96%
Rover Camera	50	46	4	92%
Mobile IP Camera	50	43	7	86%

Here, Table I illustrates the detection capabilities of three camera systems—Computer-integrated CCTV Cameras, Rover Cameras, and Mobile IP Cameras—at varying distances for anomalous objects. The detection performance

was evaluated by placing an anomalous object at distances ranging from 0.1 to 5 meters. The Computer-integrated CCTV and Rover cameras consistently achieved instant detection up to 3.5 meters, with minor delays observed beyond 4 meters. In contrast, Mobile IP Cameras showed slightly delayed detection starting at 3 meters, with further delays at 4 meters and beyond.

Table II compares the success rates of these systems in detecting multiple anomalous objects. To evaluate this, one, two, and three objects were placed simultaneously within the cameras' coverage areas, and the systems were tested for successful detection. While Computer-integrated CCTV and Rover Cameras maintained a 100% success rate for one and two objects, the Mobile IP Camera's performance dropped to 66.67% for two objects and 33.33% for three.

Table III evaluates the overall object detection accuracy of the systems. Each camera was tested with 50 trials, and the number of correctly detected objects and failures were recorded. The results show that Computer-integrated CCTV Cameras achieved the highest accuracy of 96%, followed by Rover Cameras at 92%, and Mobile IP Cameras at 86%.

VII. COMPARATIVE ANALYSIS

A. Comparison between other research works

TABLE IV. COMPARISON BETWEEN OTHER RESEARCH WORKS

Systems	Architecture	Max number of camera used	Multiple locations monitoring ability	Accuracy
Our System	Yolov8 DCNN	5	Yes	96%
Home Service Robot[11]	DCGAN	1	No	90.72%
Citizen Surveillance[12]	YOLOv2	1	No	88%
Fast Human Detection[13]	HOG	1	No	86%
MODT[14]	Kalman Filter	1	No	76.23%
AnomalyNet[8]	LTSM	1	No	95.6%
Detect Occluded Targets [15]	YOLOv3	1	No	95.7%
Underwater Object Detection [16]	Faster R-CNN	1	No	95.68%

B. Comparison with Conventional Security System

TABLE V. COMPARISON WITH CONVENTIONAL SECURITY SYSTEM

Systems	No of Test	Detected	Failed	Accuracy(%)
Our System	50	46	4	92%
Security Guard	50	41	9	82%

The system's performance, as demonstrated in Table IV, stands out when compared to other research works. It not only supports integration with a higher number of cameras, enabling simultaneous monitoring of multiple locations but also consistently outperforms these peer systems, ensuring exceptional reliability and efficiency.

Additionally, as highlighted in Table V, the system was directly tested against security guards in detection scenarios. In this controlled environment, it surpassed human performance, further establishing its superiority and dependability in real-world applications. This evidence

underscores the advanced capabilities and trustworthiness of the system in diverse operational contexts.

VIII. COST ANALYSIS

TABLE VI. COST ANALYSIS OF THE PROJECT WITH SURVEILLANCE ROVER

Component's Name	Price (BDT)	Quantity	Total Price (BDT)
Raspberry Pi 4	15000	1	15000
Web Camera	3000	3	9000
LIDAR	20000	1	20000
Motor	1500	6	9000
Motor Driver	1600	2	3200
ESP32	575	1	575
Wheel	650	6	3900
Battery, BMS	5000	1	5000
Body	5000	1	5000
Pi Display	3500	1	3500
API and Web	10000	1	10000
Others			400
Total			84575

TABLE VII. COST ANALYSIS OF THE PROJECT WITHOUT SURVEILLANCE ROVER

Component's Name	Price (BDT)	Quantity	Total Price (BDT)
Web Camera	3000	2	6000
API and Web	10000	1	10000
Total			16000

As the system can be utilized only for security purposes without the rover, the estimated cost would be around BDT 16000 or 133\$. On the other hand, with the rover surveillance ability the whole system will cost around BDT 84575 or 700\$.

IX. CONCLUSION

In summary, the paper presents a revolutionary approach to security and surveillance for commercial infrastructures. This pioneering system introduces a sophisticated system incorporating real-time video supervision, a decision-making Deep Convolutional Neural Network (DCNN) model, and safety protocols for the swift detection and mitigation of anomalous events. The future scope for this surveillance system includes a strong expansion plan to smoothly monitor over 20 cameras at once, guaranteeing scalability, efficiency, and adaptability to various business infrastructures. In addition to the comprehensive strategy outlined for future development, the integration of IR cameras will bring a unique advantage to the security and surveillance infrastructure, particularly in scenarios where conventional cameras may struggle to provide full visibility, such as when items are hidden beneath pockets, coats, or shirts. IR cameras can effectively penetrate visual barriers, enabling the system to detect and identify objects that would otherwise remain hidden from standard camera views. Our future work aims to extend detection capabilities to recognize suspicious or anomalous behaviors, such as physical altercations or harassment in public and commercial spaces. By integrating advanced behavior analysis with our existing system, we intend to provide enhanced security measures to ensure human safety.

However, challenges such as sensitivity to weather conditions, financial constraints, and limited camera range

indicate the need for ongoing enhancements. The project envisions a promising future scope marked by scalability. Ultimately, the paper makes a significant contribution to security systems through its innovative technology, emphasis on social responsibility, and commitment to continual development. The potential improvements in safety, efficiency, and sustainability across diverse commercial environments underscore its significance as a noteworthy advancement in the field of security and surveillance.

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